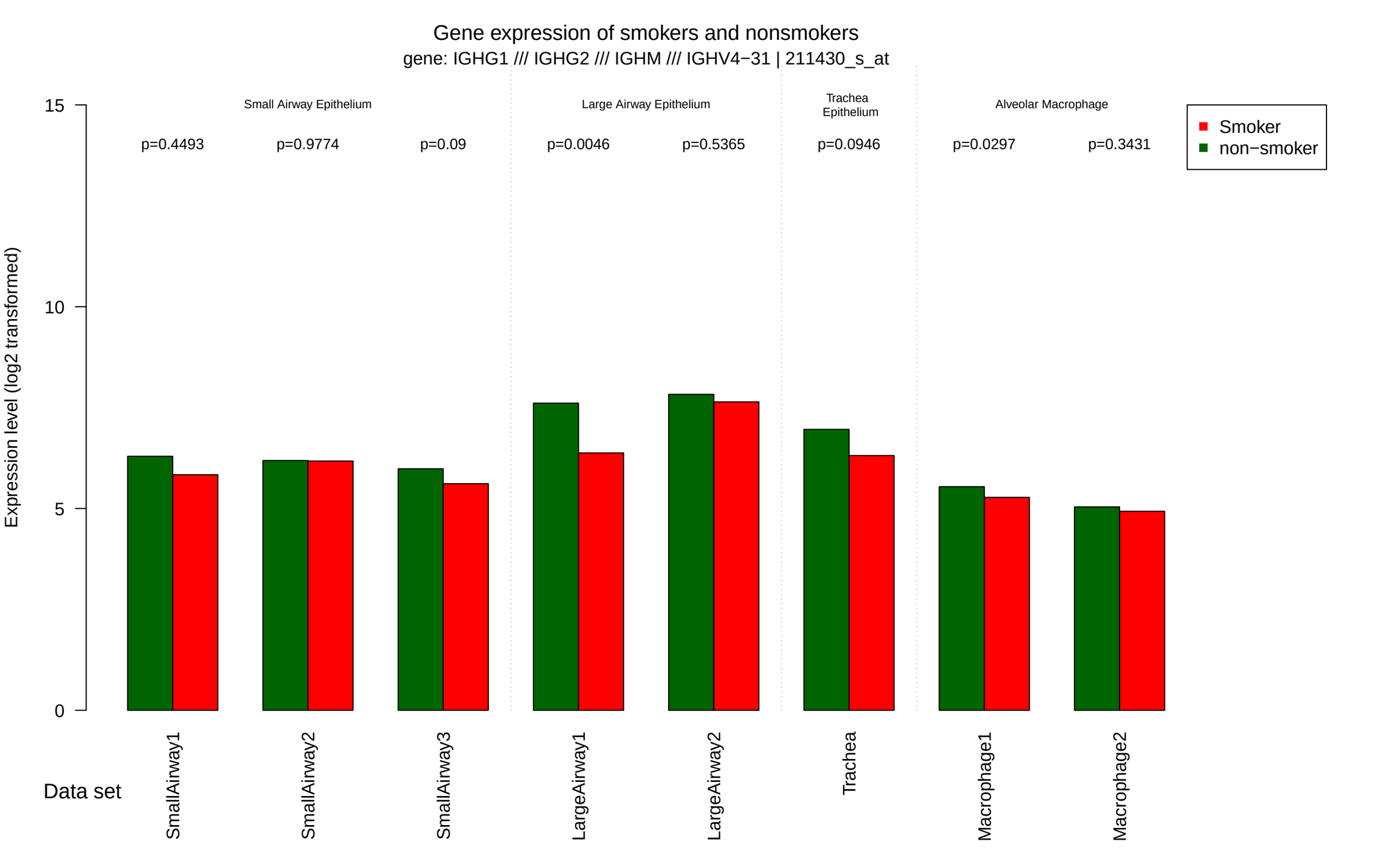
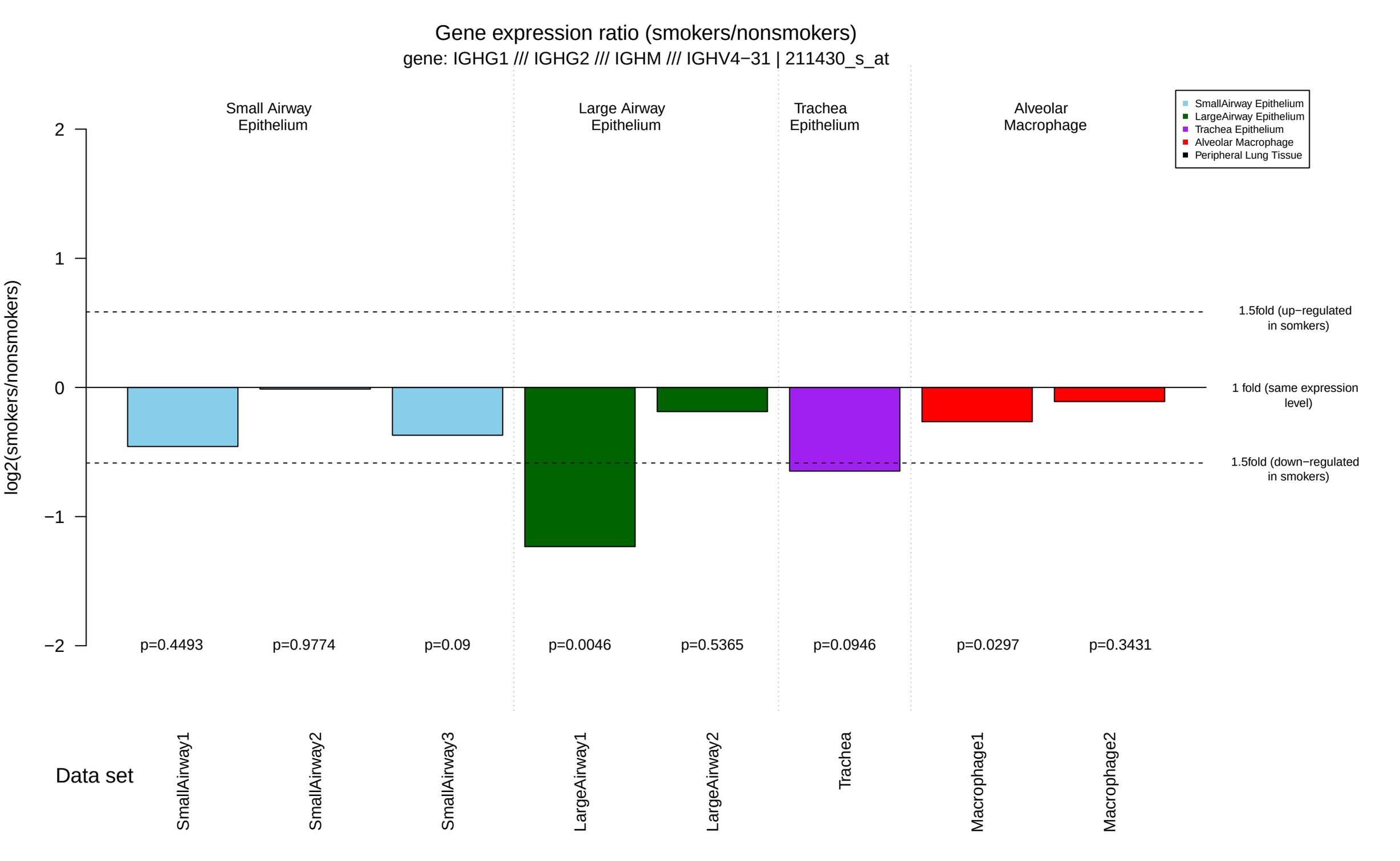
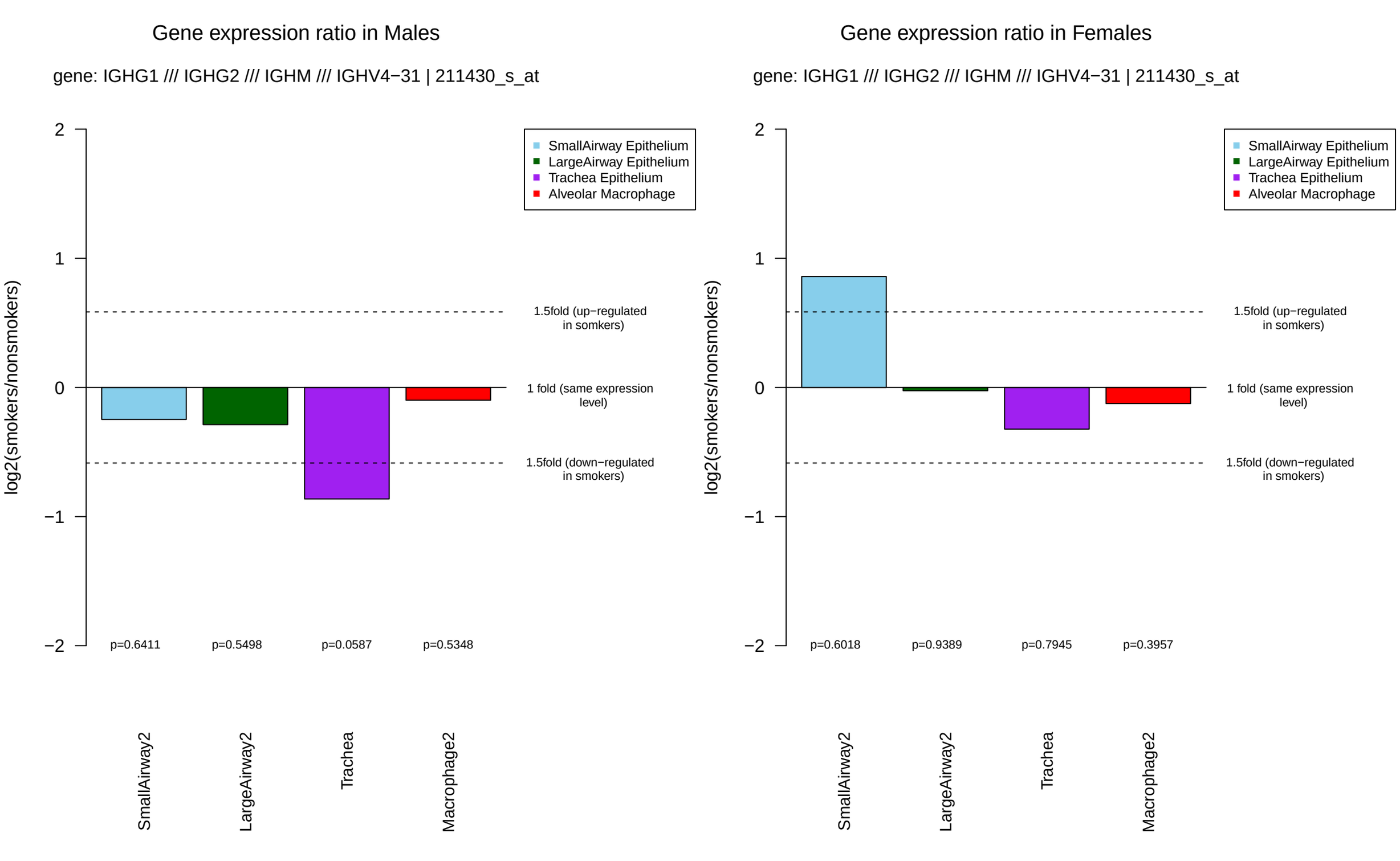
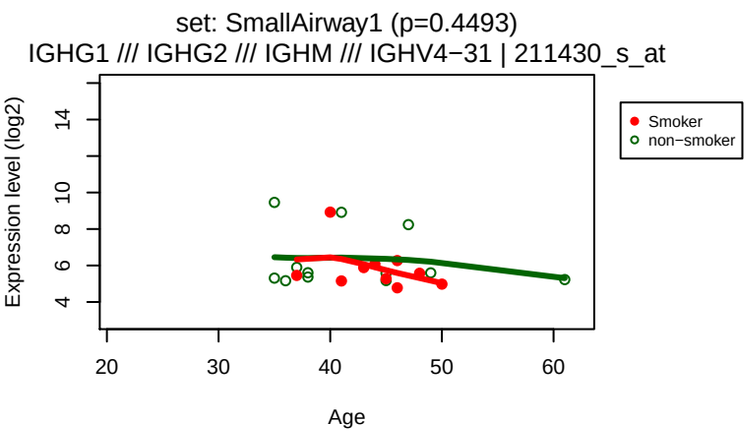
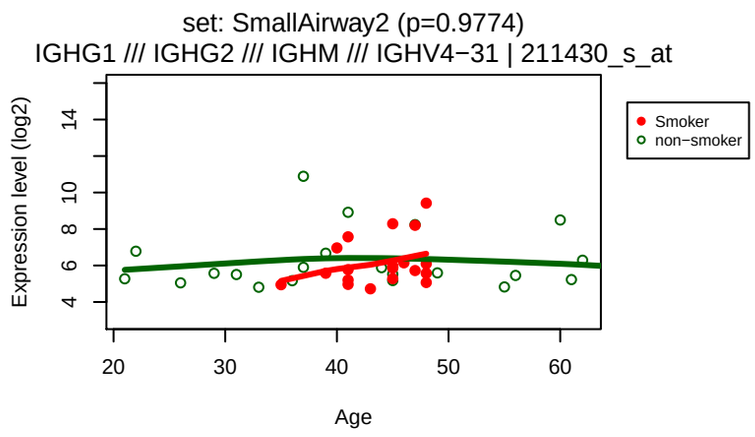
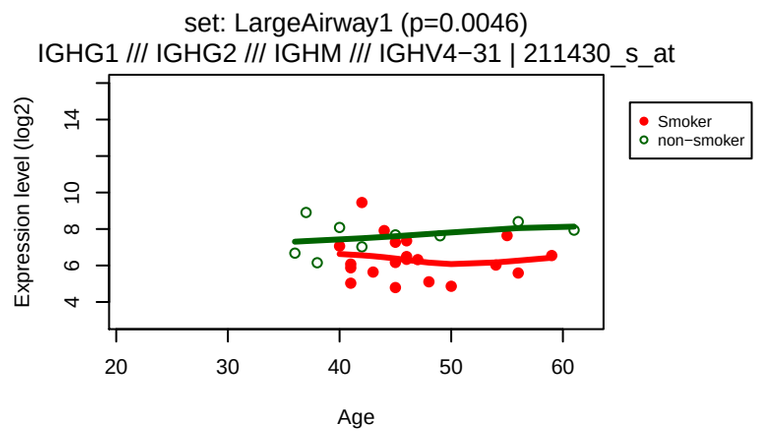

Gene expression of smokers and nonsmokers
gene: IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4493
p=0.9774
p=0.09
p=0.0046
p=0.5365
p=0.0946
p=0.0297
p=0.3431
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

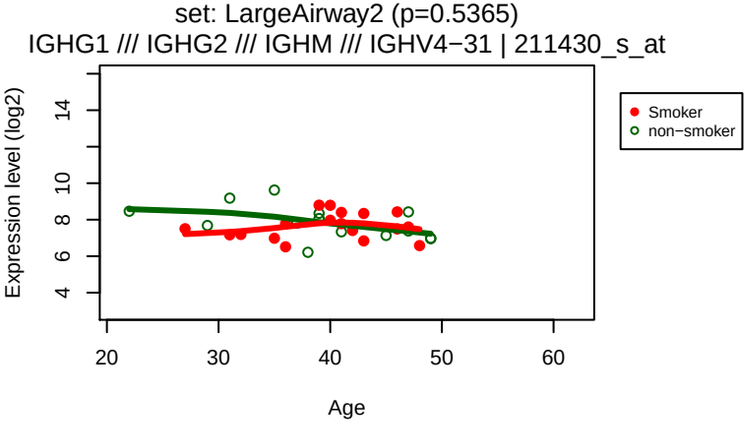

Gene expression ratio (smokers/nonsmokers)
gene: IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4493
p=0.9774
p=0.09
p=0.0046
p=0.5365
p=0.0946
p=0.0297
p=0.3431
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

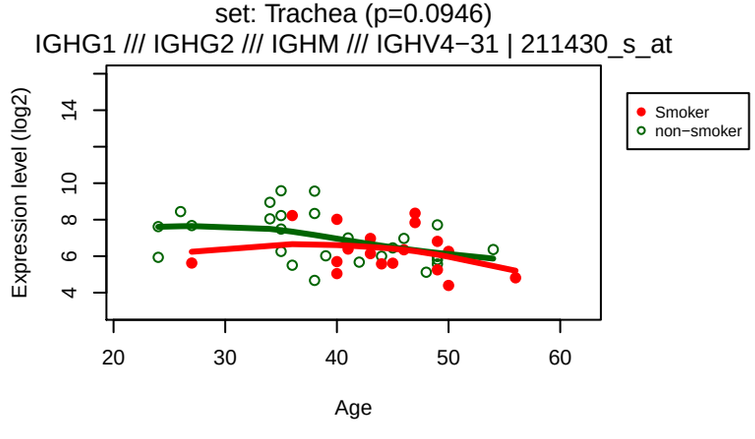

Gene expression ratio in Males
gene: IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6411
p=0.5498
p=0.0587
p=0.5348
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6018
p=0.9389
p=0.7945
p=0.3957
SmallAirway2
LargeAirway2
Trachea
Macrophage2

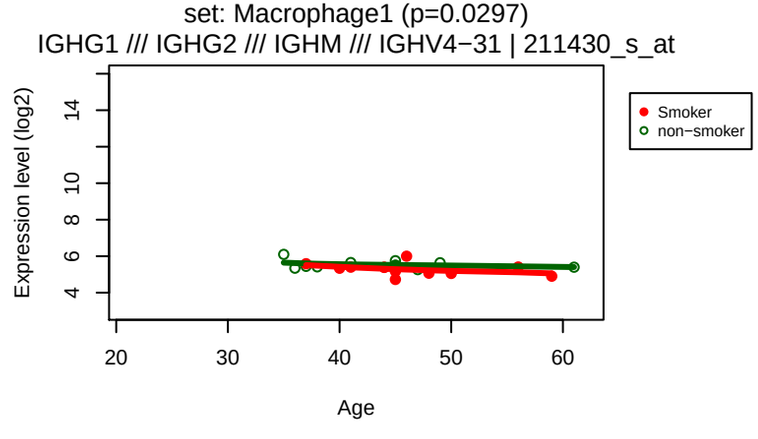

set: SmallAirway1 (p=0.4493)
IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Expression level (log2)
Age
Smoker
non−smoker

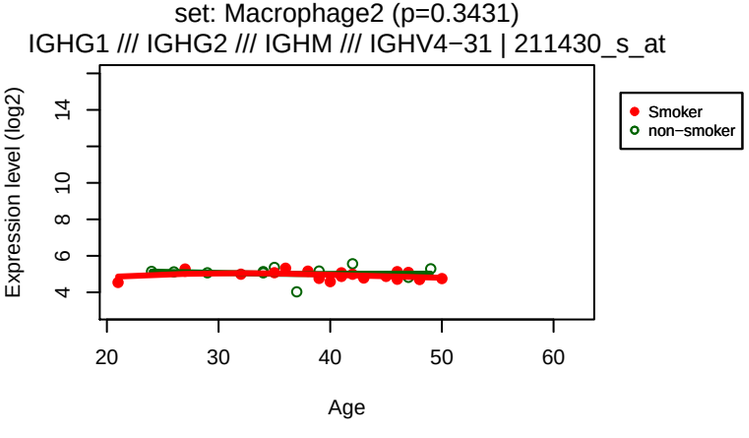

set: SmallAirway2 (p=0.9774)
IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Expression level (log2)
Age
Smoker
non−smoker

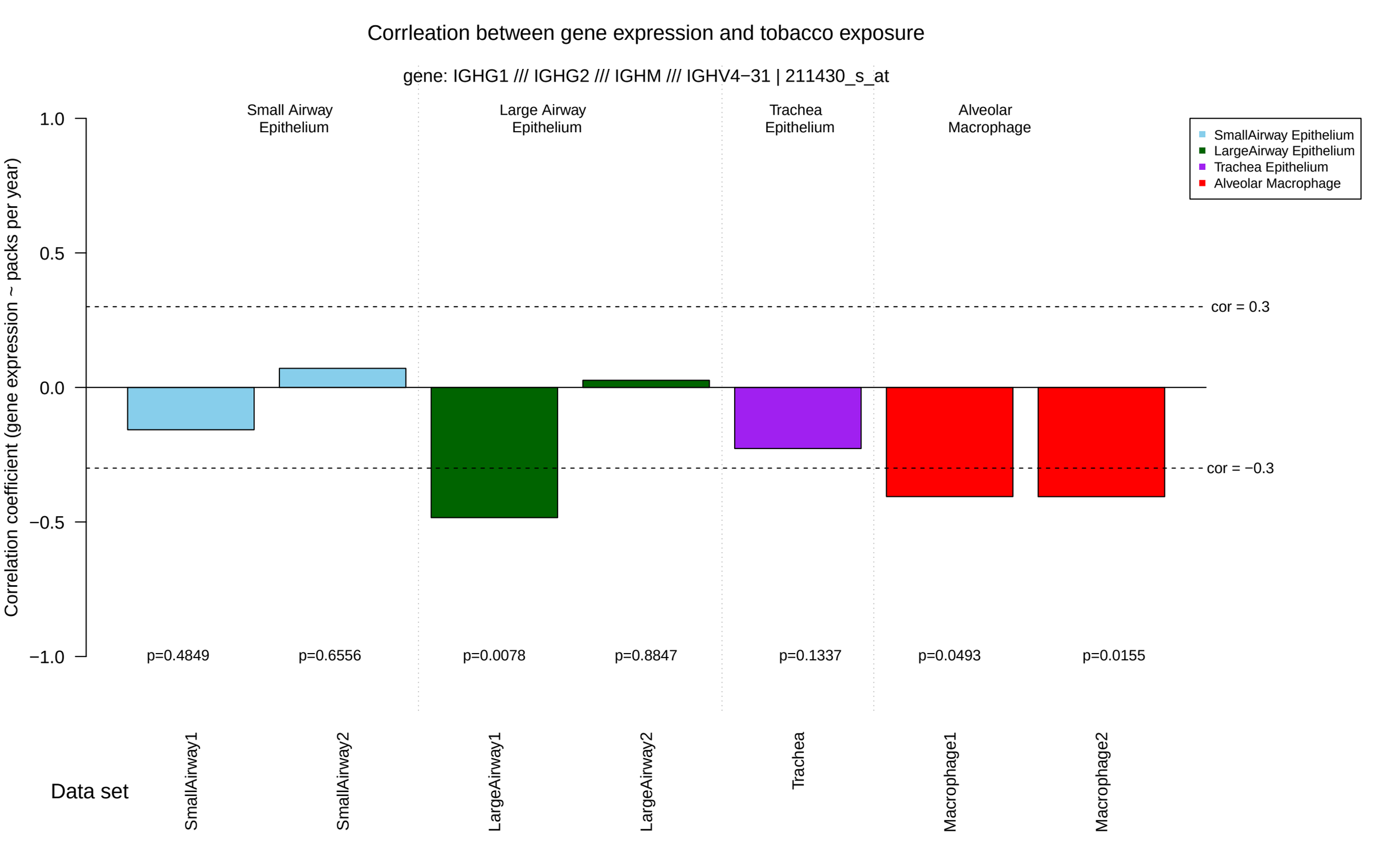

set: LargeAirway1 (p=0.0046)
IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5365)
IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0946)
IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0297)
IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3431)
IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHG1 /// IGHG2 /// IGHM /// IGHV4−31 | 211430_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4849
p=0.6556
p=0.0078
p=0.8847
p=0.1337
p=0.0493
p=0.0155
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2